

12.3: Laplace's Equation in Rectangular Coordinates

The temperature $u = u(x, y, t)$ in a two-dimensional plate satisfies the two-dimensional heat equation

$$u_t = a^2(u_{xx} + u_{yy}), \quad (12.3.1)$$

where (x, y) varies over the interior of the plate and $t > 0$. To find a solution of Equation 12.3.1, it is necessary to specify the initial temperature $u(x, y, 0)$ and conditions that must be satisfied on the boundary. However, as $t \rightarrow \infty$, the influence of the initial condition decays, so

$$\lim_{t \rightarrow \infty} u_t(x, y, t) = 0$$

and the temperature approaches a steady state distribution $u = u(x, y)$ that satisfies

$$u_{xx} + u_{yy} = 0. \quad (12.3.2)$$

This is **Laplace's equation**. This equation also arises in applications to fluid mechanics and potential theory; in fact, it is also called **the potential equation**. We seek solutions of Equation 12.3.2 in a region R that satisfy specified conditions – called **boundary conditions** – on the boundary of R . For example, we may require u to assume prescribed values on the boundary. This is called a **Dirichlet condition**, and the problem is called a **Dirichlet problem**. Or, we may require the normal derivative of u at each point (x, y) on the boundary to assume prescribed values. This is called a **Neumann condition**, and the problem is called a **Neumann problem**. In some problems we impose Dirichlet conditions on part of the boundary and Neumann conditions on the rest. Then we say that the boundary conditions and the problem are **mixed**.

Solving boundary value problems for Equation 12.3.2 over general regions is beyond the scope of this book, so we consider only very simple regions. We begin by considering the rectangular region shown in Figure 12.3.1 .

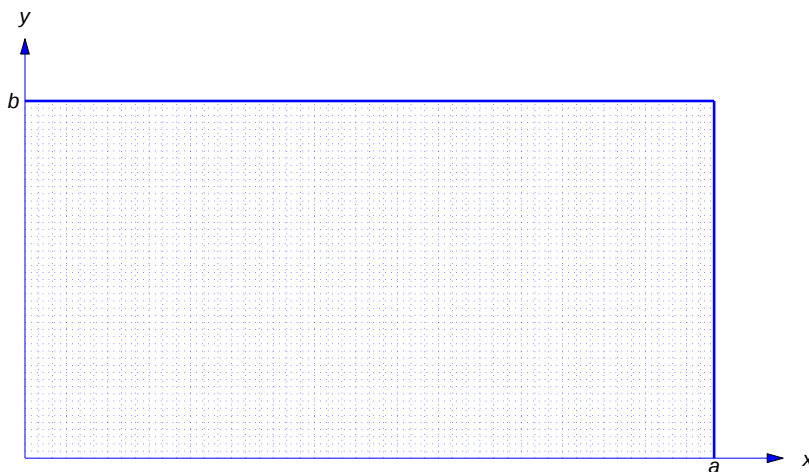


Figure 12.3.1 : A rectangular region and its boundary

The possible boundary conditions for this region can be written as

$$\begin{aligned} (1 - \alpha)u(x, 0) + \alpha u_y(x, 0) &= f_0(x), & 0 \leq x \leq a, \\ (1 - \beta)u(x, b) + \beta u_y(x, b) &= f_1(x), & 0 \leq x \leq a, \\ (1 - \gamma)u(0, y) + \gamma u_x(0, y) &= g_0(y), & 0 \leq y \leq b, \\ (1 - \delta)u(a, y) + \delta u_x(a, y) &= g_1(y), & 0 \leq y \leq b, \end{aligned}$$

where α, β, γ , and δ can each be either 0 or 1; thus, there are 16 possibilities. Let $\text{BVP}(\alpha, \beta, \gamma, \delta)(f_0, f_1, g_0, g_1)$ denote the problem of finding a solution of Equation 12.3.2 that satisfies these conditions. This is a Dirichlet problem if

$$\alpha = \beta = \gamma = \delta = 0$$

(see Figure 12.3.2), or a Neumann problem if

$$\alpha = \beta = \gamma = \delta = 1$$

(see Figure 12.3.3). The other 14 problems are mixed.

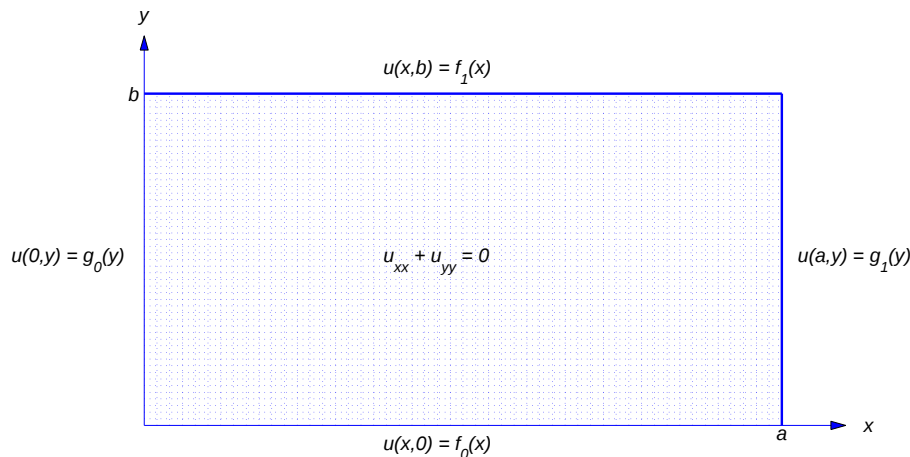


Figure 12.3.2 : A Dirichlet problem

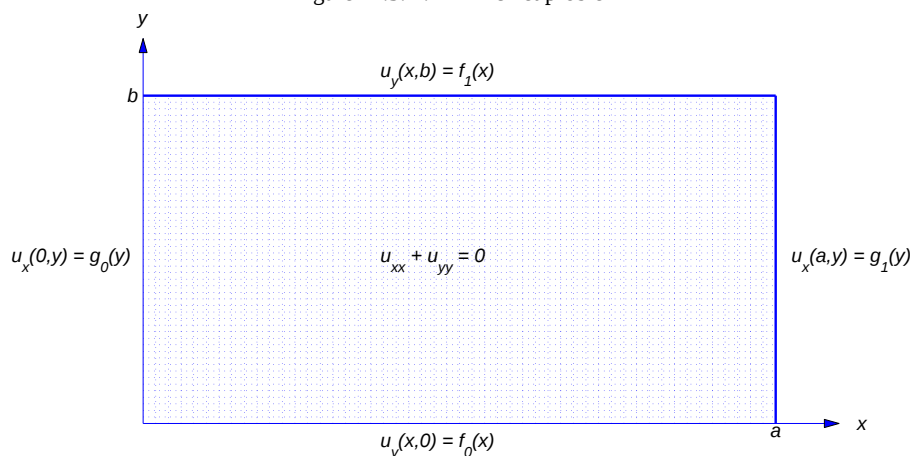


Figure 12.3.3 : A Neumann problem

For given $(\alpha, \beta, \gamma, \delta)$, the sum of solutions of

$$\begin{aligned} & \text{BVP}(\alpha, \beta, \gamma, \delta)(f_0, 0, 0, 0), \quad \text{BVP}(\alpha, \beta, \gamma, \delta)(0, f_1, 0, 0), \\ & \text{BVP}(\alpha, \beta, \gamma, \delta)(0, 0, g_0, 0), \quad \text{and} \quad \text{BVP}(\alpha, \beta, \gamma, \delta)(0, 0, 0, g_1) \end{aligned}$$

is a solution of

$$\text{BVP}(\alpha, \beta, \gamma, \delta)(f_0, f_1, g_0, g_1).$$

Therefore we concentrate on problems where only one of the functions f_0, f_1, g_0, g_1 isn't identically zero. There are 64 (count them!) problems of this form. Each has homogeneous boundary conditions on three sides of the rectangle, and a nonhomogeneous boundary condition on the fourth. We use separation of variables to find infinitely many functions that satisfy Laplace's equation and the three homogeneous boundary conditions in the open rectangle. We then use these solutions as building blocks to construct a formal solution of Laplace's equation that also satisfies the nonhomogeneous boundary condition. Since it is not feasible to consider all 64 cases, we'll restrict our attention in the text to just four. Others are discussed in the exercises.

If $v(x, y) = X(x)Y(y)$ then

$$v_{xx} + v_{yy} = X''Y + XY'' = 0$$

for all (x, y) if and only if

$$\frac{X''}{X} = -\frac{Y''}{Y} = k$$

for all (x, y) , where k is a separation constant. This equation is equivalent to

$$X'' - kX = 0, \quad Y'' + kY = 0. \quad (12.3.3)$$

From here, the strategy depends upon the boundary conditions. We illustrate this by examples.

✓ Example 12.3.1

Define the formal solution of

$$\begin{aligned} u_{xx} + u_{yy} &= 0, & 0 < x < a, & \quad 0 < y < b, \\ u(x, 0) &= f(x), & u(x, b) &= 0, & \quad 0 \leq x \leq a, \\ u(0, y) &= 0, & u(a, y) &= 0, & \quad 0 \leq y \leq b \end{aligned} \quad (12.3.4)$$

(Figure 12.3.4).

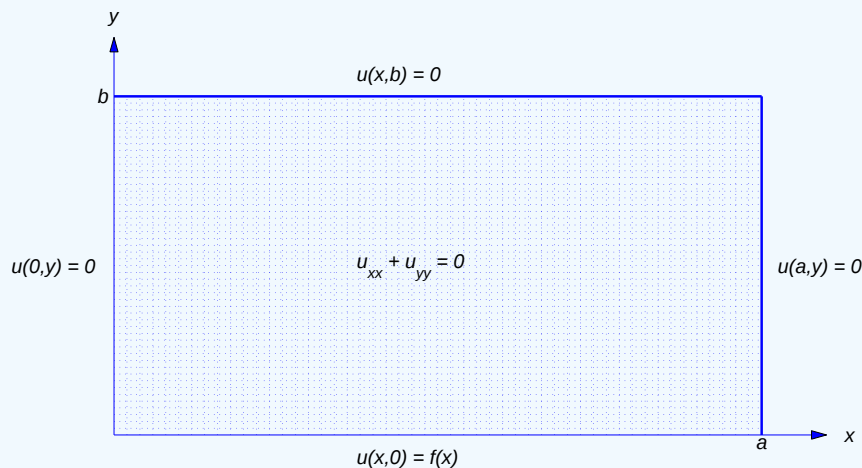


Figure 12.3.4 : The boundary value problem 12.3.4

Solution:

The boundary conditions in Equation 12.3.4 require products $v(x, y) = X(x)Y(y)$ such that $X(0) = X(a) = Y(b) = 0$; hence, we let $k = -\lambda$ in Equation 12.3.3. Thus, X and Y must satisfy

$$X'' + \lambda X = 0, \quad X(0) = 0, \quad X(a) = 0 \quad (12.3.5)$$

and

$$Y'' - \lambda Y = 0, \quad Y(b) = 0. \quad (12.3.6)$$

From Theorem 11.1.2, the eigenvalues of Equation 12.3.5 are $\lambda_n = n^2\pi^2/a^2$, with associated eigenfunctions

$$X_n = \sin \frac{n\pi x}{a}, \quad n = 1, 2, 3, \dots$$

Substituting $\lambda = n^2\pi^2/a^2$ into Equation 12.3.6 yields

$$Y'' - (n^2\pi^2/a^2)Y = 0, \quad Y(b) = 0,$$

so we could take

$$Y_n = \sinh \frac{n\pi(b-y)}{a}; \quad (12.3.7)$$

however, because of the nonhomogeneous Dirichlet condition at $y = 0$, it is better to require that $Y_n(0) = 1$, which can be achieved by dividing the right side of Equation 12.3.7 by its value at $y = 0$; thus, we take

$$Y_n = \frac{\sinh n\pi(b-y)/a}{\sinh n\pi b/a}.$$

Then

$$v_n(x, y) = X_n(x)Y_n(y) = \frac{\sinh n\pi(b-y)/a}{\sinh n\pi b/a} \sin \frac{n\pi x}{a},$$

so $v_n(x, 0) = \sin n\pi x/a$ and v_n satisfies Equation 12.3.4 with $f(x) = \sin n\pi x/a$. More generally, if $\alpha_1, \dots, \alpha_m$ are arbitrary constants then

$$u_m(x, y) = \sum_{n=1}^m \alpha_n \frac{\sinh n\pi(b-y)/a}{\sinh n\pi b/a} \sin \frac{n\pi x}{a}$$

satisfies Equation 12.3.4 with

$$f(x) = \sum_{n=1}^m \alpha_n \sin \frac{n\pi x}{L}.$$

Therefore, if f is an arbitrary piecewise smooth function on $[0, a]$, we define the formal solution of Equation 12.3.4 to be

$$u(x, y) = \sum_{n=1}^{\infty} \alpha_n \frac{\sinh n\pi(b-y)/a}{\sinh n\pi b/a} \sin \frac{n\pi x}{a}, \quad (12.3.8)$$

where

$$S(x) = \sum_{n=1}^{\infty} \alpha_n \sin \frac{n\pi x}{a}$$

is the Fourier sine series of f on $[0, a]$; that is,

$$\alpha_n = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi x}{a} dx, \quad n = 1, 2, 3, \dots$$

If $y < b$ then

$$\frac{\sinh n\pi(b-y)/a}{\sinh n\pi b/a} \approx e^{-n\pi y/a} \quad (12.3.9)$$

for large n , so the series in Equation 12.3.8 converges if $0 < y < b$; moreover, since also

$$\frac{\cosh n\pi(b-y)/a}{\sinh n\pi b/a} \approx e^{-n\pi y/a}$$

for large n , Theorem 12.1.2 applied twice with $z = x$ and twice with $z = t$, shows that u_{xx} and u_{yy} can be obtained by differentiating u term by term if $0 < y < b$. **Exercise 12.3.37.** Therefore u satisfies Laplace's equation in the interior of the rectangle in Figure 12.3.4. Moreover, the series in 12.3.8 also converges on the boundary of the rectangle, and satisfies the three homogeneous boundary conditions in 12.3.4. Therefore, since $u(x, 0) = S(x)$ for $0 \leq x \leq L$, u is an actual solution of 12.3.5 if and only if $S(x) = f(x)$ for $0 \leq x \leq a$. From Theorem 11.3.2, this is true if f is continuous and piecewise smooth on $[0, L]$, and $f(0) = f(L) = 0$.

✓ Example 12.3.2

Solve Equation 12.3.4 with $f(x) = x(x^2 - 3ax + 2a^2)$.

Solution

From Example 11.3.6,

$$S(x) = \frac{12a^3}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{n^3} \sin \frac{n\pi x}{a}.$$

Therefore

$$u(x, y) = \frac{12a^3}{\pi^3} \sum_{n=1}^{\infty} \frac{\sinh n\pi(b-y)/a}{n^3 \sinh n\pi b/a} \sin \frac{n\pi x}{a}. \quad (12.3.10)$$

To compute approximate values of $u(x, y)$, we must use partial sums of the form

$$u_m(x, y) = \frac{12a^3}{\pi^3} \sum_{n=1}^m \frac{\sinh n\pi(b-y)/a}{n^3 \sinh n\pi b/a} \sin \frac{n\pi x}{a}.$$

Because of Equation 12.3.9, small values of m provide sufficient accuracy for most applications if $0 < y < b$. Moreover, the n^3 in the denominator in 12.3.10 ensures that this is also true for $y = 0$. For graphing purposes, we chose $a = 2, b = 1$, and $m = 10$. Figure 12.3.5 shows the surface

$$u = u(x, y), \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 1,$$

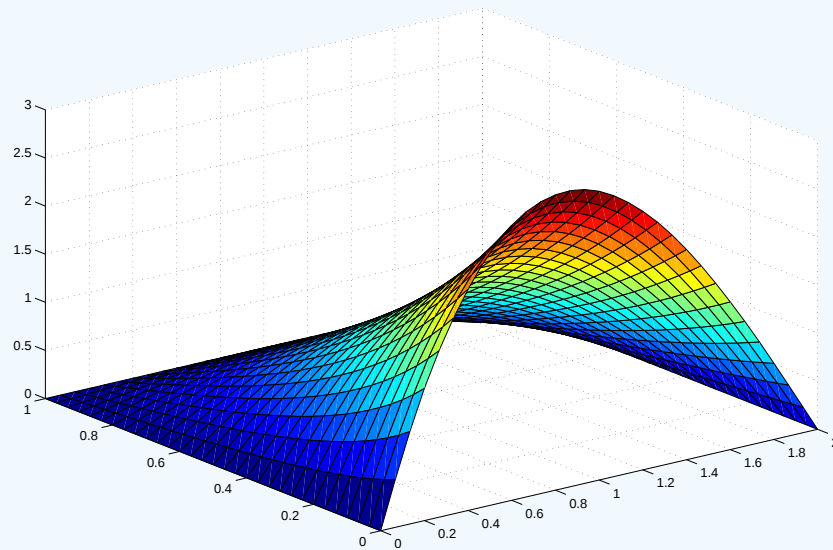


Figure 12.3.5

while Figure 12.3.6 shows the curves

$$u = u(x, 0.1k), \quad 0 \leq x \leq 2, \quad k = 0, 1, \dots, 10.$$

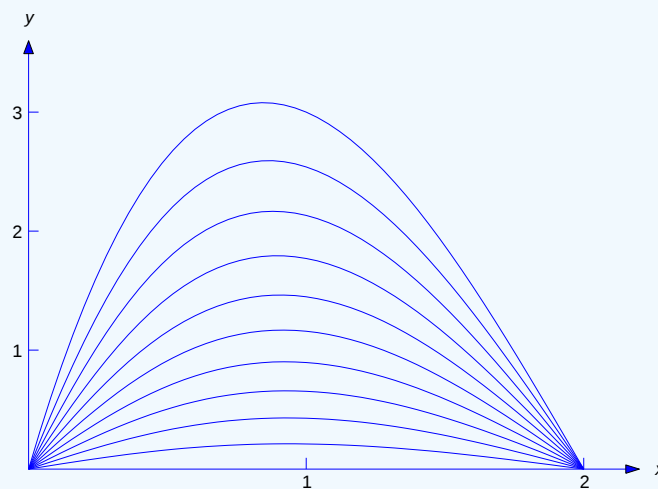


Figure 12.3.6

✓ Example 12.3.3

Define the formal solution of

$$\begin{aligned} u_{xx} + u_{yy} &= 0, & 0 < x < a, & \quad 0 < y < b, \\ u(x, 0) &= 0, & u_y(x, b) &= f(x), & 0 \leq x \leq a, \\ u_x(0, y) &= 0, & u_x(a, y) &= 0, & 0 \leq y \leq b \end{aligned} \tag{12.3.11}$$

(Figure 12.3.7).

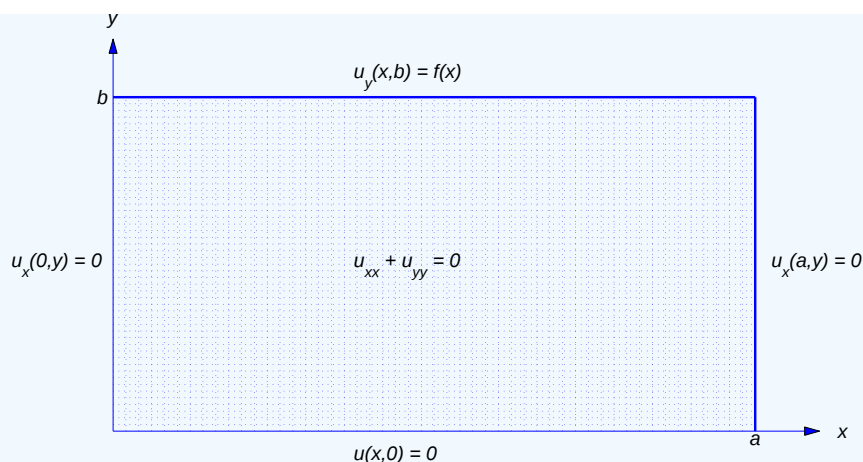


Figure 12.3.7 : The boundary value problem 12.3.11

Solution

The boundary conditions in Equation 12.3.11 require products $v(x, y) = X(x)Y(y)$ such that $X'(0) = X'(a) = Y(0) = 0$; hence, we let $k = -\lambda$ in Equation 12.3.3. Thus, X and Y must satisfy

$$X'' + \lambda X = 0, \quad X'(0) = 0, \quad X'(a) = 0 \quad (12.3.12)$$

and

$$Y'' - \lambda Y = 0, \quad Y(0) = 0. \quad (12.3.13)$$

From Theorem 11.1.3, the eigenvalues of Equation 12.3.12 are $\lambda = 0$, with associated eigenfunction $X_0 = 1$, and $\lambda_n = n^2\pi^2/a^2$, with associated eigenfunctions

$$X_n = \cos \frac{n\pi x}{a}, \quad n = 1, 2, 3, \dots$$

Since $Y_0 = y$ satisfies Equation 12.3.13 with $\lambda = 0$, we take $v_0(x, y) = X_0(x)Y_0(y) = y$. Substituting $\lambda = n^2\pi^2/a^2$ into Equation 12.3.13 yields

$$Y'' - (n^2\pi^2/a^2)Y = 0, \quad Y(0) = 0,$$

so we could take

$$Y_n = \sinh \frac{n\pi y}{a}. \quad (12.3.14)$$

However, because of the nonhomogeneous Neumann condition at $y = b$, it is better to require that $Y'_n(b) = 1$, which can be achieved by dividing the right side of Equation 12.3.14 by the value of its derivative at $y = b$; thus,

$$Y_n = \frac{a \sinh n\pi y/a}{n\pi \cosh n\pi b/a}.$$

Then

$$v_n(x, y) = X_n(x)Y_n(y) = \frac{a \sinh n\pi y/a}{n\pi \cosh n\pi b/a} \cos \frac{n\pi x}{a},$$

so

$$\frac{\partial v_n}{\partial y}(x, b) = \cos \frac{n\pi x}{a}.$$

Therefore v_n satisfies Equation 12.3.11 with $f(x) = \cos n\pi x/a$. More generally, if $\alpha_0, \dots, \alpha_m$ are arbitrary constants then

$$u_m(x, y) = \alpha_0 y + \frac{a}{\pi} \sum_{n=1}^m \alpha_n \frac{\sinh n\pi y/a}{n \cosh n\pi b/a} \cos \frac{n\pi x}{a}$$

satisfies Equation 12.3.11 with

$$f(x) = \alpha_0 + \sum_{n=1}^m \alpha_n \cos \frac{n\pi x}{L}.$$

Therefore, if f is an arbitrary piecewise smooth function on $[0, a]$ we define the formal solution of Equation 12.3.11 to be

$$u(x, y) = \alpha_0 y + \frac{a}{\pi} \sum_{n=1}^{\infty} \alpha_n \frac{\sinh n\pi y/a}{n \cosh n\pi b/a} \cos \frac{n\pi x}{a},$$

where

$$C(x) = \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos \frac{n\pi x}{a}$$

is the Fourier cosine series of f on $[0, a]$; that is,

$$\alpha_0 = \frac{1}{a} \int_0^a f(x) dx \quad \text{and} \quad \alpha_n = \frac{2}{a} \int_0^a f(x) \cos \frac{n\pi x}{a} dx, \quad n = 1, 2, 3, \dots$$

✓ Example 12.3.4

Solve Equation 12.3.11 with $f(x) = x$.

Solution

From Example 11.3.1,

$$C(x) = \frac{a}{2} - \frac{4a}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi x}{a}$$

Therefore

$$u(x, y) = \frac{ay}{2} - \frac{4a^2}{\pi^3} \sum_{n=1}^{\infty} \frac{\sinh(2n-1)\pi y/a}{(2n-1)^3 \cosh(2n-1)\pi b/a} \cos \frac{(2n-1)\pi x}{a} \quad (12.3.15)$$

For graphing purposes, we chose $a = 2$, $b = 1$, and retained the terms through $n = 10$ in Equation 12.3.15. Figure 12.3.8 shows the surface

$$u = u(x, y), \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 1,$$

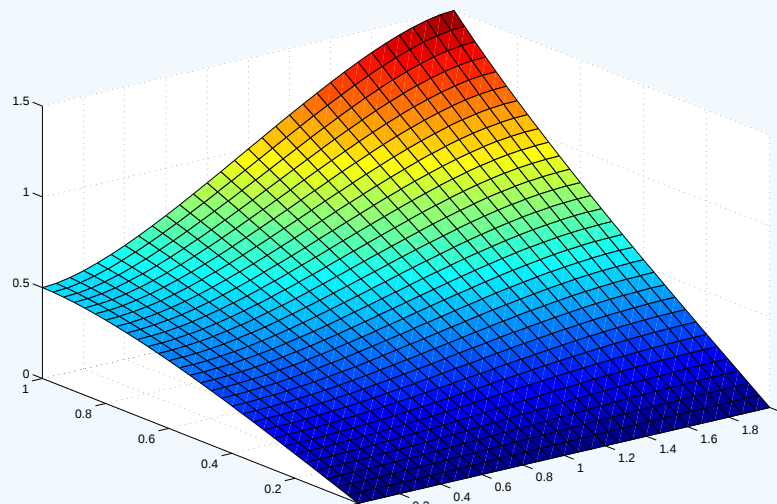


Figure 12.3.8

while Figure 12.3.9 shows the curves

$$u = u(x, .1k), \quad 0 \leq x \leq 2, \quad k = 0, 1, \dots, 10.$$

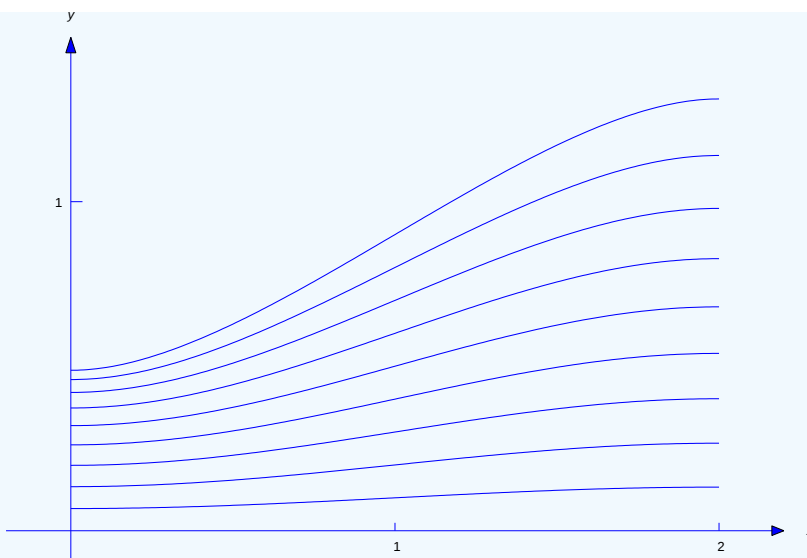


Figure 12.3.9

✓ Example 12.3.5

Define the formal solution of

$$\begin{aligned} u_{xx} + u_{yy} &= 0, & 0 < x < a, & \quad 0 < y < b, \\ u(x, 0) &= 0, & u_y(x, b) &= 0, & \quad 0 \leq x \leq a, \\ u(0, y) &= g(y), & u_x(a, y) &= 0, & \quad 0 \leq y \leq b \end{aligned} \quad (12.3.16)$$

(Figure 12.3.10).

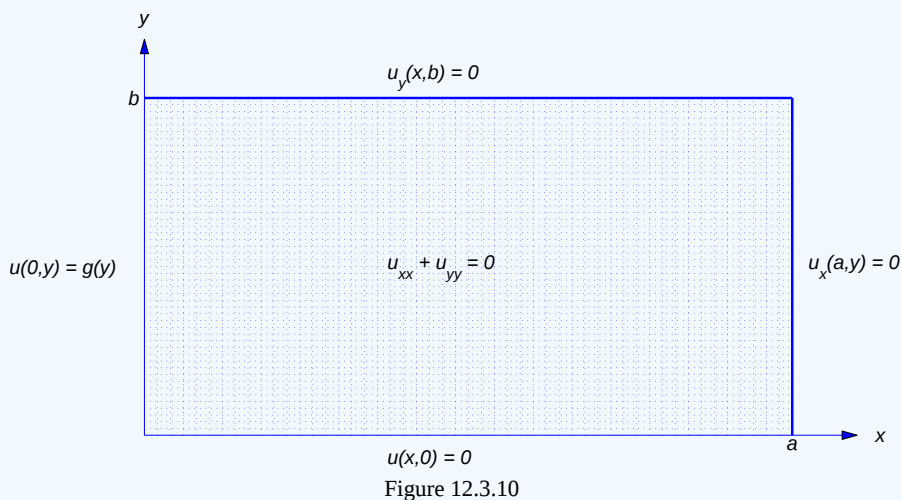


Figure 12.3.10

Solution

The boundary conditions in Equation 12.3.16 require products $v(x, y) = X(x)Y(y)$ such that $Y(0) = Y'(b) = X'(a) = 0$; hence, we let $k = \lambda$ in Equation 12.3.3. Thus, X and Y must satisfy

$$X'' - \lambda X = 0, \quad X'(a) = 0 \quad (12.3.17)$$

and

$$Y'' + \lambda Y = 0, \quad Y(0) = 0, \quad Y'(b) = 0. \quad (12.3.18)$$

From Theorem 11.1.4, the eigenvalues of Equation 12.3.18 are $\lambda_n = (2n-1)^2\pi^2/4b^2$, with associated eigenfunctions

$$Y_n = \sin \frac{(2n-1)\pi y}{2b}, \quad n = 1, 2, 3, \dots$$

Substituting $\lambda = (2n - 1)^2\pi^2/4b^2$ into Equation 12.3.17 yields

$$X'' - ((2n - 1)^2\pi^2/4b^2)X = 0, \quad X'(a) = 0,$$

so we could take

$$X_n = \cosh \frac{(2n - 1)\pi(x - a)}{2b}. \quad (12.3.19)$$

However, because of the nonhomogeneous Dirichlet condition at $x = 0$, it is better to require that $X_n(0) = 1$, which can be achieved by dividing the right side of Equation 12.3.19 by its value at $x = 0$; thus,

$$X_n = \frac{\cosh(2n - 1)\pi(x - a)/2b}{\cosh(2n - 1)\pi a/2b}.$$

Then

$$v_n(x, y) = X_n(x)Y_n(y) = \frac{\cosh(2n - 1)\pi(x - a)/2b}{\cosh(2n - 1)\pi a/2b} \sin \frac{(2n - 1)\pi y}{2b},$$

so

$$v_n(0, y) = \sin \frac{(2n - 1)\pi y}{2b}.$$

Therefore v_n satisfies Equation 12.3.16 with $g(y) = \sin(2n - 1)\pi y/2b$. More generally, if $\alpha_1, \dots, \alpha_m$ are arbitrary constants then

$$u_m(x, y) = \sum_{n=1}^m \alpha_n \frac{\cosh(2n - 1)\pi(x - a)/2b}{\cosh(2n - 1)\pi a/2b} \sin \frac{(2n - 1)\pi y}{2b}$$

satisfies Equation 12.3.16 with

$$g(y) = \sum_{n=1}^m \alpha_n \sin \frac{(2n - 1)\pi y}{2b}.$$

Thus, if g is an arbitrary piecewise smooth function on $[0, b]$, we define the formal solution of Equation 12.3.16 to be

$$u(x, y) = \sum_{n=1}^{\infty} \alpha_n \frac{\cosh(2n - 1)\pi(x - a)/2b}{\cosh(2n - 1)\pi a/2b} \sin \frac{(2n - 1)\pi y}{2b},$$

where

$$S_M(x) = \sum_{n=1}^{\infty} \alpha_n \sin \frac{(2n - 1)\pi y}{2b}$$

is the mixed Fourier sine series of g on $[0, b]$; that is,

$$\alpha_n = \frac{2}{b} \int_0^b g(y) \sin \frac{(2n - 1)\pi y}{2b} dy.$$

✓ Example 12.3.6

Solve Equation 12.3.16 with $g(y) = y(2y^2 - 9by + 12b^2)$.

Solution

From Example 11.3.8,

$$S_M(y) = \frac{96b^3}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n - 1)^3} \left[3 + (-1)^n \frac{4}{(2n - 1)\pi} \right] \sin \frac{(2n - 1)\pi y}{2b}$$

Therefore

$$u(x, y) = \frac{96b^3}{\pi^3} \sum_{n=1}^{\infty} \frac{\cosh(2n - 1)\pi(x - a)/2b}{\cosh(2n - 1)\pi a/2b} \left[3 + (-1)^n \frac{4}{(2n - 1)\pi} \right] \sin \frac{(2n - 1)\pi y}{2b}$$

✓ Example 12.3.7

Define the formal solution of

$$\begin{aligned} u_{xx} + u_{yy} &= 0, & 0 < x < a, & \quad 0 < y < b, \\ u_y(x, 0) &= 0, & u(x, b) &= 0, & \quad 0 \leq x \leq a, \\ u_x(0, y) &= 0, & u_x(a, y) &= g(y), & \quad 0 \leq y \leq b \end{aligned} \quad (12.3.20)$$

(Figure 12.3.11).

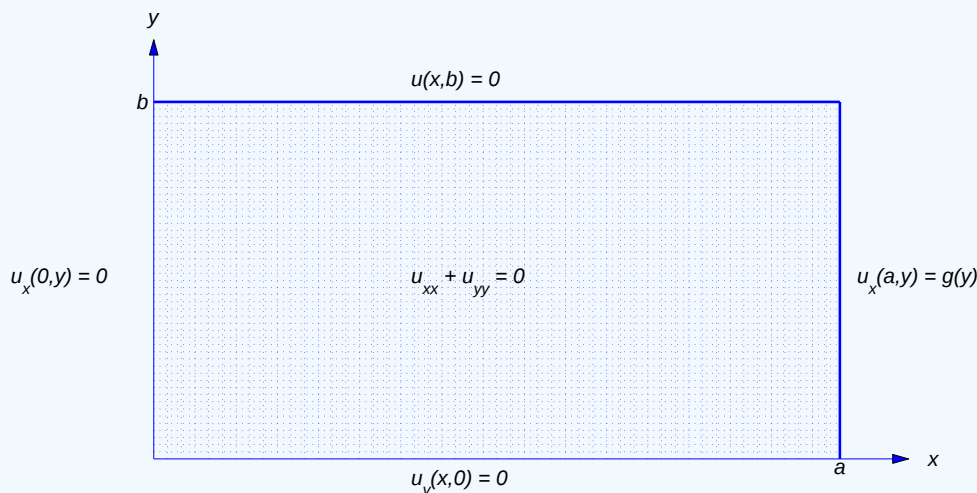


Figure 12.3.11 : The boundary value problem 12.3.20

Solution:

The boundary conditions in Equation 12.3.20 require products $v(x, y) = X(x)Y(y)$ such that $Y'(0) = Y(b) = X'(0) = 0$; hence, we let $k = \lambda$ in Equation 12.3.3. Thus, X and Y must satisfy

$$X'' - \lambda X = 0, \quad X'(0) = 0 \quad (12.3.21)$$

and

$$Y'' + \lambda Y = 0, \quad Y'(0) = 0, \quad Y(b) = 0. \quad (12.3.22)$$

From Theorem 11.1.4, the eigenvalues of Equation 12.3.22 are $\lambda_n = (2n-1)^2\pi^2/4b^2$, with associated eigenfunctions

$$Y_n = \cos \frac{(2n-1)\pi y}{2b}, \quad n = 1, 2, 3, \dots$$

Substituting $\lambda = (2n-1)^2\pi^2/4b^2$ into Equation 12.3.21 yields

$$X'' - ((2n-1)^2\pi^2/4b^2)X = 0, \quad X'(0) = 0,$$

so we could take

$$X_n = \cosh \frac{(2n-1)\pi x}{2b}. \quad (12.3.23)$$

However, because of the nonhomogeneous Neumann condition at $x = a$, it is better to require that $X'_n(a) = 1$, which can be achieved by dividing the right side of Equation 12.3.23 by the value of its derivative at $x = a$; thus,

$$X_n = \frac{2b \cosh(2n-1)\pi x/2b}{(2n-1)\pi \sinh(2n-1)\pi a/2b}.$$

Then

$$v_n(x, y) = X_n(x)Y_n(y) = \frac{2b \cosh(2n-1)\pi x/2b}{(2n-1)\pi \sinh(2n-1)\pi a/2b} \cos \frac{(2n-1)\pi y}{2b},$$

so

$$\frac{\partial v_n}{\partial x}(a, y) = \cos \frac{(2n-1)\pi y}{2b}.$$

Therefore v_n satisfies Equation 12.3.20 with $g(y) = \cos(2n-1)\pi y/2b$. More generally, if $\alpha_1, \dots, \alpha_m$ are arbitrary constants then

$$u_m(x, y) = \frac{2b}{\pi} \sum_{n=1}^m \alpha_n \frac{\cosh(2n-1)\pi x/2b}{(2n-1) \sinh(2n-1)\pi a/2b} \cos \frac{(2n-1)\pi y}{2b}$$

satisfies Equation 12.3.20 with

$$g(y) = \sum_{n=1}^{\infty} \alpha_n \cos \frac{(2n-1)\pi y}{2b}.$$

Therefore, if g is an arbitrary piecewise smooth function on $[0, b]$, we define the formal solution of Equation 12.3.20 to be

$$u(x, y) = \frac{2b}{\pi} \sum_{n=1}^{\infty} \alpha_n \frac{\cosh(2n-1)\pi x/2b}{(2n-1) \sinh(2n-1)\pi a/2b} \cos \frac{(2n-1)\pi y}{2b},$$

where

$$C_M(y) = \sum_{n=1}^{\infty} \alpha_n \cos \frac{(2n-1)\pi y}{2b}$$

is the mixed Fourier cosine series of g on $[0, b]$; that is,

$$\alpha_n = \frac{2}{b} \int_0^b g(y) \cos \frac{(2n-1)\pi y}{2b} dy.$$

✓ Example 12.3.8

Solve Equation 12.3.20 with $g(y) = y - b$.

Solution

From Example 12.3.3,

$$C_M(y) = -\frac{8b}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi y}{2b}.$$

Therefore

$$u(x, y) = -\frac{16b^2}{\pi^3} \sum_{n=1}^{\infty} \frac{\cosh(2n-1)\pi x/2b}{(2n-1)^3 \sinh(2n-1)\pi a/2b} \cos \frac{(2n-1)\pi y}{2b}.$$

Laplace's Equation for a Semi-Infinite Strip

We now seek solutions of Laplace's equation on the semi-infinite strip

$$S: \{0 < x < a, \quad y > 0\}$$

that satisfy homogeneous boundary conditions at $x = 0$ and $x = a$, and a nonhomogeneous Dirichlet or Neumann condition at $y = 0$ (Figure 12.3.12). An example of such a problem is

$$\begin{aligned} u_{xx} + u_{yy} &= 0 & 0 < x < a, \quad y > 0 \\ u(x, 0) &= f(x) & 0 \leq x \leq a \\ u(0, y) &= 0 & y > 0 \\ u(a, y) &= 0 & y > 0 \end{aligned}$$

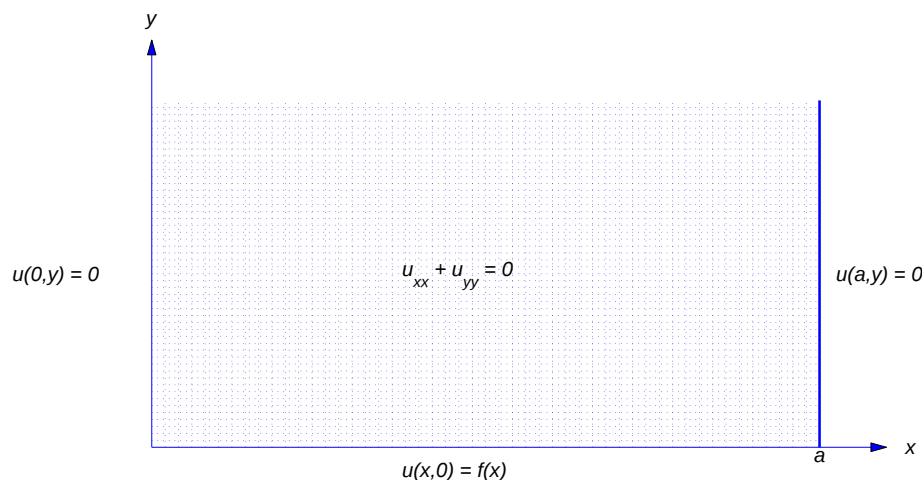


Figure 12.3.12 : A boundary value problem on a semi-infinite strip

The boundary conditions in this problem are not sufficient to determine u , for if $u_0 = u_0(x, y)$ is a solution and K is a constant then

$$u_1(x, y) = u_0(x, y) + K \sin \frac{\pi x}{a} \sinh \frac{\pi y}{a}.$$

is also a solution. (Verify.) However, if we also require — on physical grounds — that the solution remain bounded for all (x, y) in S then $K = 0$ and this difficulty is eliminated.

✓ Example 12.3.9

Define the bounded formal solution of Equation ???.

Solution

Proceeding as in the solution of Example 12.3.1, we find that the building block functions are of the form

$$v_n(x, y) = Y_n(y) \sin \frac{n\pi x}{a},$$

where

$$Y_n'' - (n^2\pi^2/a^2)Y_n = 0.$$

Therefore

$$Y_n = c_1 e^{n\pi y/a} + c_2 e^{-n\pi y/a}$$

where c_1 and c_2 are constants. Although the boundary conditions in Equation ??? don't restrict c_1 , and c_2 , we must set $c_1 = 0$ to ensure that Y_n is bounded. Letting $c_2 = 1$ yields

$$v_n(x, y) = e^{-n\pi y/a} \sin \frac{n\pi x}{a},$$

and we define the bounded formal solution of Equation ??? to be

$$u(x, y) = \sum_{n=1}^{\infty} b_n e^{-n\pi y/a} \sin \frac{n\pi x}{a},$$

where

$$S(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{a}$$

is the Fourier sine series of f on $[0, a]$.

See **Exercises 12.3.29-12.3.34** for other boundary value problems on a semi-infinite strip.

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